Week 5 web page: delivery guide

Instructor notes for Week5_SimulatingData.html

General

- Page: https://kevintshoemaker.github.io/NRES-746/Week5_SimulatingData.html. Open it before class; the three widgets need no internet once loaded.
- For live demos, run the matching section of `CODE_Week5_SimulatingData.R` (same seeds, same numbers). Everything runs in a few seconds.
- Zoom the browser to 125-150% for the projector.
- Questions on the page have no answers. Answers are below; *Land:* is where discussion should end up.
- Readings for Canvas: Bolker 5.1-5.2 (printed pp. 197-208) and the lognormal, 4.5.2.6 (pp. 178-180). Section 5.3 (power) goes with Week 6.
- WODs: Monday *Choosing a Data-Generating Model* (key in `wod/answer_keys/`). Wednesday *Writing the Likelihood* replaced the lecture.

As taught (Sept. 21-23, 2026)

- **Monday** (per the announcement, added Nov. 4): opened by reviewing population mean and variance and deriving $E[\bar{X}] = \mu$ (unbiased) and $\text{Var}(\bar{X}) = \sigma^2/n$ under i.i.d. (now sharpened in the Week 4 page's `#wod-sample-mean`), then Bolker ch. 5 data-generating models, ending with the WOD. Its worked example was a negative exponential mean with gamma noise, scale = mean/shape; that example is now in `#anatomy` (seed 1002) and the script.
- **Lab 4 opening:** no report. The page follows the plans in the two PDF NOTES (now in `lecture_notes/archive/`).
- **Lab 4:** only Act 1 was handed out. Act 2 (the power analysis) became the Week 6 Wednesday live-coding demo; Act 3 was never used.
- **Wednesday:** the instructor was away. *Writing the Likelihood* ran as a group exercise in place of the lecture and was debriefed the next Monday (see the Week 6 guide).
- The page was built Oct. 29, after the week, from those NOTES and the legacy LECTURE3 page (now in `archive/`). New on the page: the three widgets, the geometric tree model (from LECTURE3's unshown code), and the lek null simulation.

Part 1: Simulating data from a model (Monday, ~35 min)

Big picture (~2 min)

- Point at MODELS. Forward (parameters to data) vs. backward (data to parameters, from Wednesday on). Ask for reasons to simulate before showing the list.

WOD (~8 min; runs long)

- Walk through data set 1 on the page (exponential decline, negative binomial, $a = 55$, $b = 0.06$, $k = 8$).
- "For data sets 2 and 3, which function would you use for the expected value, and which distribution for the noise? What feature of each data set rules out a normal distribution?" *Land:* nest fate, logistic expected value with Bernoulli noise (0/1 outcomes; a probability must stay between 0 and 1). Chick mass, Michaelis-Menten (truth $350x/(5 + x)$) with gamma noise (truth shape 25), lognormal also fine (positive, and the spread grows with the mean).

Random number generators (~2 min; skim if behind)

- `r` + parameters; `d`, `p`, `q` in one sentence. The Beta(0.1, 0.1) draws print as 0 and 1 because they're extreme, not because of a bug.

Signal and noise (~8 min)

- Bolker's two examples. The point: the function sets the *mean* of the distribution; "plus noise" only works for the normal.
- Negative exponential + gamma: the mean isn't a gamma parameter, so set scale = μ_i/s (mean = shape × scale); s plays the role of σ (bigger s, less scatter).
- Widget, in order: Exponential decline with Normal (spread cards about 2 at both ends), Poisson (about 5 at low x, 3-4 at high x), Lognormal (about 7, then 3-4). Numbers move with each data set.
- "Choose 'Exponential decline' and compare Normal, Poisson, and Lognormal noise using the two spread cards. Which keeps the scatter the same everywhere, and which let it shrink as the mean shrinks? Then choose 'Linear,' drag b below about -1, and choose Poisson. Why do some points disappear?" *Land:* normal keeps it constant; Poisson (variance = mean) and lognormal (SD proportional to mean) shrink it. A linear mean goes negative, and a Poisson mean can't be negative, so those points can't be simulated (the warning says how many). Same lesson with Logistic + Bernoulli vs. Linear + Bernoulli.

Many replicates (~7 min)

- Straight line: every tree below the model's expectation; misses biggest at girths 5-7. Geometric model (volume $\propto$ girth^3, lognormal) fits, girth 7 at the low edge.

- "Every red point also falls inside the boxes of the no-relationship model. Does that make it as good a model of these trees as the geometric one? What would a sharper comparison need to account for?" *Land:* no. It "fits" by spreading probability over everything, including negative volumes, so it would fit almost any data. A sharper comparison rewards a model for putting probability where the data actually are; that is the likelihood (Week 6). Plausible is not the same as corroborated.

Null simulation (~5 min)

- Eight years: slope -3.87 males per year, $p = 0.071$. Ten years: $p = 0.011$. Stress that the null had to specify the noise too.

Discussion (~3 min)

- Q1 ("A manager asks whether the lek 'started declining in year 8.' What would you tell them?") *Land:* the counts were simulated with a decline from year 0. Eight years couldn't separate it from noise ($p \approx 0.07$); not ruling out "no trend" isn't evidence of no trend, and the p-value doesn't date the decline. Low power.
- Q2 (assumed less scatter than the real counts) *Land:* null slopes too tightly bunched, so the p-value is too small: false alarms. Too much assumed scatter gives the reverse.
- Q3 (own data) open. Steer toward simulating from each candidate and comparing with the real data, as in section 5.

Bonus: process vs. observation noise (widget)

- $\sigma = 0.15$: year-30 SD about 0.15 (observation) vs. about 0.8 (process; $\sigma\sqrt{30} = 0.82$). Same draws, different place.
- "With $\sigma = 0.15$, compare the year-30 spread in the two panels. A straight line fit to one process-noise run treats its 30 points as independent scatter around a line. Why might its confidence interval for the slope be too narrow?" *Land:* process deviations carry over, so neighboring years are correlated; 30 correlated points hold less information than 30 independent ones, the regression's SE is too small, and the interval too narrow (the coverage widget shows about 24% for a nominal 90%).
- This is the noise argument from the Lab 4 Act 1 scenario (its index noise is observation noise).

Part 2: Trends on the log scale (Lab 4 opening, ~20 min)

- Don't give away Act 1 before students try it: λ and b for a 25% decline over 25 years (1a), the raw/log sketches (1b), which OLS assumption the raw scale breaks (1c), whether every-other-year surveys change the center or spread of $\hat{b}$ (2a-2b), the Monte Carlo SE formula $\sqrt{p(1-p)/R}$ and the $R \geq 900$ answer (2c), the simulation plan (Part 3), and $\text{SD}(\hat{b}) = \sigma/\sqrt{S_{tt}}$ (4C). The page uses a 40% decline over 20 years and never shows the replicate loop.

- **Lognormal (~4 min):** median vs. mean, CV constant (0.151 at $\sigma = 0.15$); SD 6.10 at median 40, 3.66 at median 24.
- **Log-linear fit (~6 min):** $\lambda = 0.9748$, $b = -0.0255$; seed-2201 fit $\hat{b} = -0.0299$, $\hat{\lambda} = 0.9706$, estimated decline 0.45 (truth 0.40). Back-transform gives the median.
- **Coverage (~7 min):** one interval $(-0.0395, -0.0202)$ contains the truth. Widget defaults (20 years, $\sigma = 0.15$, 90%, observation noise).
  - "Simulate 100 programs a few times with observation noise. How close to 90% does the coverage come, and how much does it change from one batch of 100 to the next? Then switch to process noise. What happens, and why?" *Land:* about 90% over many batches, but a batch of 100 easily lands anywhere from about 84% to 96% (that's Monte Carlo error; let Act 1 2c quantify it). Process noise: about 23-25%, for the reason in the bonus. More years doesn't fix it (about 38% at 10 years, 21% at 30).
  - Avoid using the years slider to talk about precision before Act 1 Part 2 (it previews 2a).
- **Error rates and power (~3 min):** definitions, the directional rule, and a pointer to the Week 6 page. Act 1 doesn't need more.
- Act 1 key: `labs/answer_keys/Lab4_Act1_Key.pdf`.

Part 3: Writing a likelihood (Wednesday)

- WOD-only day with the instructor away. The page section only points to the Week 6 page's `#wod-likelihood`; delivery notes are in the Week 6 guide.